



09.12.2023 (E)

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 55	Class: SY	Semester: III (SVU 2020)
Programme: Honours in Cyber Security and Forensics		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer Engineering
Course Code: 116h55C301	Name of the Course: Applied Cryptography	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define Confidentiality. Discuss attacks on confidentiality.	5
ii)	The Euler's totient function $[\Phi(n)]$ gives number of numbers those are co-prime with n. If n is expressed as $n=p*q$, where p and q are prime, $\Phi(n)$ can be computed as $\Phi(n) = \Phi(p) * \Phi(q)$. A. Discuss which one of the algebraic homomorphic property this relationship holds. B. Discuss cryptographic significance of this algebraic <u>homomorphic property(not the mentioned relation)</u> .	2+3
iii)	Encrypt the message "The lone wolf dies, but the pack survives" using: a. Playfair cipher with key: Encipherment b. Rail fence cipher with fence height=4	5
iv)	Explain in brief any two attacks on RSA	5
v)	Explain the concept of 'Cryptanalysis' with suitable example	5
vi)	What is man-in-middle attack? How it can be prevented?	2+3

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Compare and contrast Feistel cipher and non- Feistel ciphers	5
ii)	Explain Key generation process in DES with a suitable diagram.	5
	OR	
Q2 A	In a cryptosystem, a sender is confused with which one of the following messages she should send to the receiver. $M_1 = \{E_{K-Private-Sender}[E_{K-Public-Receiver}[D]] + E_{K-shared}[\text{Hash}[D]] + \text{One_time_pad} + \text{Time_stamp}\}$ $M_2 = \{E_{K-Public-Receiver}[E_{K-Private-Sender}[D]] + E_{K-shared}[\text{Hash}[D]] + \text{Random_number_challenge} + \text{Time_stamp}\}$ Where M=Message, D= Data, E=Encryption, K=Key. A. Discuss in brief which one of the messages should be preferred and Why? B. <u>List</u> and explain in 1-2 lines which one of the security goals are achieved by the chosen message? C. Does the chosen message solve problems of non-repudiation on sender's-receiver's side and replay attack? Discuss the same.	2+4+4
Q 2 B	Solve any One	10
i)	Explain the following terms and their cryptographic significance:	10

	a. Key stream Bias b. Forward security c. Avalanche effect d. Initialization vector	
ii)	Compare and Contrast Symmetric Key and Asymmetric Key Cryptography	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	In a Elliptic Curve cryptographic system defined over $E_{23}(1,1)$, the two points on the curve are given as $P=(3,10)$ and $Q=(9,7)$. Compute $P+Q$.	10
ii)	Explain Working of RC4 encryption algorithm. List any two weaknesses, too.	10
iii)	Discuss how symmetric and asymmetric key encryption can be used for message authentication. Clearly state if the chosen method achieves confidentiality and authentication.	10
	[Represent contents of communication clearly those give authentication; you may refer message contents given Q. 2 A for the same.] M_1 & M_2 .	

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain in brief working of SHA-512 with a suitable diagram.	10
ii)	Explain how "Secure Multiparty Computing" is a privacy-preserving computation method? Discuss if homomorphic encryption and zero knowledge proofs can be treated as cryptographic alternatives to Secure Multiparty Computing?	10
iii)	Discuss the Quantum key Distribution process. What are limitations to Quantum key cryptography in current scenario?	6+4

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Write a short note on security attacks.	5
ii)	Discuss strengths and weaknesses in AES design	5
iii)	Write a short note on : Applications of Digital Certificates	5
iv)	Explain cryptographic obfuscation. List any two limitations of cryptographic obfuscation.	3+2
v)	Discuss contents of digital certificates as per X.509 V3 format	5
vi)	Discuss strengths, Limitations and Challenges in ECC	5



Semester: July 2023 --October 2023		
Maximum Marks:50	Examination: ESE Examination	Duration:2 Hrs.
Programme code: 54		
Programme: B. Tech. (Honors in Data Science & Analytics)	Class: S.Y.	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116H54C301	Name of the Course: Data Visualization	
Instructions: 1) Ensure you are copy pasting maps/ graphs in the word file answer sheet template 2) Save the word file as well as the PDF file of the answer sheet template. 3) Assume suitable data wherever necessary and justify the same		

Que. No	Question	Max. Mark
Q1	Solve any TWO	10
i)	What do you understand by primary data and secondary data?	5
ii)	Explain advantaged and limitations of 4 point Likert scales.	5
iii)	Explain giving examples what are "mark cards" with reference to Tableau software	5
iv)	Explain exploratory and explanatory analysis.	5

Use AirLinesData.csv file shared in the folder to solve question no. 2 and 3

Que. No.	Question	Max. Marks
Q2 A	Solve	20
i)	Use any excel functions to remove blank rows, columns, remove duplicate entries from the sample data set shared Save the modified data as "cleanData.xls"	10
ii)	Use appropriate graphs/ visualization used to (Any TWO) a) Display counts of flights per Airline or Source/Destination. b) Trends in flight counts over Date_of_Journey (e.g., daily, monthly trends). c) Distribution of flight durations or total stops.	5 5 5

Que. No.	Question	Max. Marks
Q 3	Solve	20
i)	Prepare a report/ presentations/ story to understand flights & trends, time related analysis and routes & destination based of the Airlines dataset shared.	10
ii)	Use appropriate graphs/ visualization used to (Any TWO) a) Understand distribution of Departure and Arrival times over the day/ month b) Visualise routes for source and destination c) Analyzing frequent Additional_Info terms	5 5 5



Semester: July 2023 –October 2023		
Maximum Marks:50	Examination: ESE Examination	Duration:2 Hrs.
Programme code: 54		
Programme: B. Tech. ((Honors in Data Science & Analytics))	Class: S.Y.	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116H54C301	Name of the Course: Data Visualization	
Instructions: 1) Ensure you are copy pasting maps/ graphs in the word file answer sheet template 2) Save the word file as well as the PDF file of the answer sheet template. 3) Assume suitable data wherever necessary and justify the same		

Que. No	Question	Max. Mark	CO	BT
Q1	Solve any TWO	10		
i)	Explain Gestalt principles of design.	5	CO 1, CO 2, CO 3, CO 4	UN (2)
ii)	Mention advantages and limitations of a Pie charts used for presenting data.	5		RE (1)
iii)	With reference to Tableau software explain the terms dimensions and measures	5		UN (2)
iv)	Justify why audience analysis is essential in story telling	5		AN (4)

Use HR_Analytics.csv dataset shared in the folder to solve question no. 2 and 3

Que. No.	Question	Max. Marks
Q2 A	Solve	20
i)	Use any excel functions to remove blank rows, columns, remove duplicate entries from the sample data set shared Save the modified data as "cleanData.xls"	10
ii)	Use appropriate graphs/ visualization used to (Any TWO) a) Analyse Attrition (e.g., how many employees left and stayed). b) Analyse of employees age-wise c) Analyse of employees salary –wise	5 5 5

Que. No.	Question	Max. Marks
Q 3	Solve	20
i)	Prepare a report/ presentations/ story to understand Job-related factors, employee demographics, attrition analysis.	10
ii)	Use appropriate graphs/ visualization used to (Any TWO) a) Analyse of employees Job-level based b) Analyse of employees Department-wise gender based c) Analyse job role-wise education filed	5 5 5



SOMAIYA
VIDYAVIHAR UNIVERSITY

08.12.2023 (E)

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 55	Class:	Semester: V (SVU 2020)
Programme: Honors in Cyber Security & Forensics (COMP)	T.Y.B.Tech	
Name of the Constituent College:		Name of the department: Computer Engineering.
K. J. Somaiya College of Engineering		
Course Code: 116h55C501	Name of the Course: Blockchain Technology	
Instructions: 1) Draw neat diagrams. 2) All questions are compulsory.		
3) Assume suitable data wherever necessary.		

Que. No.	Question	Max. Marks
Q1	Attempt <i>any Four</i> of the following:	20
i)	Elaborate on the elements of distributed computing.	5
ii)	What is the Nakamoto Consensus?	5
iii)	What is the role of asymmetric cryptography in blockchain?	5
iv)	What is a Decentralized Autonomous Organization (DAO)?	5
v)	List down the role of bitcoin miners.	5
vi)	Why is GIT not considered as blockchain?	5

Que. No.	Question	Max. Marks
Q2 A	Attempt the following:	10
i)	Write five points of difference between hard fork and soft fork.	5
ii)	Distinguish between tokens and coins.	5
	OR	
Q2 A	Explain how tampering and impersonification can be guarded using asymmetric cryptography & digital signatures.	10
Q 2 B	Attempt <i>any One</i> of the following:	10
i)	(a) What is Sybil attack in blockchain? Explain. (b) Discuss the solution to Sybil attack.	05 05
ii)	What is a consensus protocol? Explain any one consensus mechanism with example.	10

Que. No.	Question	Max. Marks
Q3	Attempt <i>any Two</i> of the following:	20
i)	Define a smart contract. Briefly explain the steps in creation of a smart contract with supporting example.	10
ii)	Justify why the Ethereum smart contract are so special compared to other programs?	10
iii)	Explain the role of Ethereum virtual machine.	10

Que. No.	Question	Max. Marks
Q4	Attempt <i>any Two</i> of the following:	20
i)	Define hyperledger fab ric. Enumerate the advantages of hyperledger.	10
ii)	What is Bitcoin mining ? How are bitcoins different from non-fungible tokens.	10
iii)	What are the different legal aspects associated with cryptocurrency exchange?	10

Que. No.	Question	Max. Marks
Q5	Write short notes on <i>any four</i> of the following:	20
i)	Merkle Patricia Tree.	5
ii)	Proof of burn.	5
iii)	Difference between Ethereum blockchain & Bitcoin.	5
iv)	Ganache.	5
v)	GHOST protocol.	5
vi)	Application of blockchain for Healthcare Management.	5



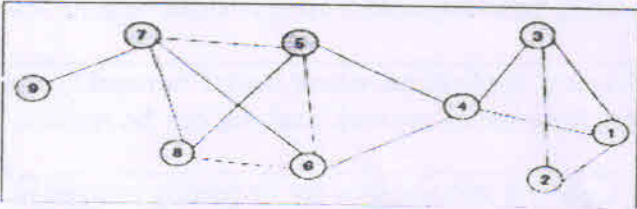
08.12.2023 (E)

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 55	Class: T.Y B.Tech	Semester:V(SVU 2020)
Programme: Honours Data Science and Analytics		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer Engineering
Course Code: 116h54C501	Name of the Course: Data Analytics	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Provide an example illustrating how diagnostics analytics can be beneficial in healthcare settings.	5
ii)	Provide example of real-world application where social network graphs are used. Discuss how insights from social network analysis can be applied in this domain.	5
iii)	What analytical software tools and technologies are frequently utilized in data analytics processes within the context of financial applications?	5
iv)	Outline the challenges associated with analyzing large datasets in the healthcare domain.	5
v)	How does the integration of sensor data benefit clinical decision-making?	5
vi)	Explain the process of querying vector data in GIS.	5

Que. No.	Question	Max. Marks																						
Q2 A	Solve the following	10																						
i)	Discuss the application of predictive analytics in marketing.	5																						
ii)	Define the Autocorrelation Test in the context of time series analysis and explain its primary purpose. How does this test help in understanding the temporal structure of a time series?	5																						
	OR																							
Q2 A	Explain the concept of overlapping communities in social network graphs. Discuss methods for finding overlapping communities and highlight the advantages of considering overlapping structures in community detection	10																						
Q 2 B	Solve any One	10																						
i)	Explain usability of raster data model in GIS for agricultural land management.	10																						
ii)	For the data given below, generate a forecast for period 11 through 13 using trend-adjusted exponential smoothing. Use $\alpha = 0.4$ and $\beta = 0.3$	10																						
	<table><tr><td>Period</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Data Values</td><td>500</td><td>524</td><td>520</td><td>528</td><td>540</td><td>542</td><td>558</td><td>550</td><td>570</td><td>575</td></tr></table>	Period	1	2	3	4	5	6	7	8	9	10	Data Values	500	524	520	528	540	542	558	550	570	575	
Period	1	2	3	4	5	6	7	8	9	10														
Data Values	500	524	520	528	540	542	558	550	570	575														

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the life cycle of Data analytics with respect to transportation	10
ii)	Discuss how Simrank is used to measure similarity between nodes in a graph. Illustrate with suitable example	10
iii)	Explain the basic steps of health care data analytics	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Consider the undirected graph below and perform the Girvan-Newman algorithm to detect communities. Provide the step-by-step process, including the calculation of edge betweenness centrality and the removal of edges 	10
ii)	Explain the role of NLP in mining clinical information	10
iii)	Compare and contrast moving average forecasts, and exponential smoothing forecasts. Discuss the strengths and limitations of each method in different time series scenarios	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	View Shed vs Water shed analysis	5
ii)	Counting triangles in graph	5
iii)	Wrangling Time series data w.r.t treating missing values and outlier data	5
iv)	Barriers in Adopting EHR	5
v)	Key functionalities of a Geographic Information System (GIS) Database Management System (DBMS)	5
vi)	Primary data sources utilized in the financial sector,	5

5.12.2023(M)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 55	Class: L.Y.B.Tech	Semester: VII (SVU 2020)
Programme: Honors in Cyber Security & Forensics (COMP).		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer Engineering.
Course Code: 116h55C701	Name of the Course: Secure Coding.	
Instructions: 1) Draw neat diagrams. 2) All questions are compulsory. 3) Assume suitable data wherever necessary.		

Que. No.	Question	Max. Marks
Q1	Attempt <i>any Four</i> of the following:	20
i)	List the elements of secure code.	5
ii)	Elaborate on the key stages of threat modelling.	5
iii)	Mention about any five tools used for code review.	5
iv)	What do you understand by cross-domain security?	5
v)	Give examples of cryptographic tools and libraries for secure coding.	5
vi)	What is sandbox security? Explain.	5

Que. No.	Question	Max. Marks
Q2 A	Attempt the following:	10
i)	Elaborate on the practices that can make passwords vulnerable.	5
ii)	List down the best practices for password security.	5
	OR	
Q2 A	Illustrate with the help of an example how C & C++ are more vulnerable to buffer overflow attack.	10
Q 2 B	Attempt <i>any One</i> of the following:	10
i)	With the help of a supporting example, discuss how malicious codes can manipulate the state of a system.	10
ii)	Classify different techniques to prevent a buffer overflow attack.	10

Que. No.	Question	Max. Marks
Q3	Attempt <i>any Two</i> of the following:	20
i)	String query = String.format ("SELECT * FROM users WHERE usr='%s' AND pwd='%s'", usr, pwd); Connection conn = db.getConn(); Statement stmt = conn.createStatement(); ResultSet rs = stmt.executeQuery(query); Discuss the outcome of the above query. Also, mention the reasons as to what leads to such attacks.	10

ii)	How do attackers exploit common security misconfigurations?	10
iii)	A major US retailer reportedly had an internal malware analysis sandbox analyzing attachments. The sandbox software had detected potentially unwanted software, but no one responded to this detection. The sandbox had been producing warnings for some time before the breach was detected due to fraudulent card transactions by an external bank. You are appointed as a consultant for investigating the above scenario. (a) Describe the possible breach that occurred and the potential reason for the same. (b) List down different guidelines and solutions in order to prevent such incidents in future.	10

Que. No.	Question	Max. Marks
Q4	Attempt any Two of the following:	20
i)	What do you understand by secure key management? List different cryptographic mistakes user commit w.r.t. key management.	10
ii)	What is verification test? Explain with the help of an example how is it different from validation test?	10
iii)	Enumerate on the different types of error in coding practices. Explain the secure coding practices for mitigating such errors.	10

Que. No.	Question	Max. Marks
Q5	Write short notes on any four of the following:	20
i)	Phases of secure-SDLC.	5
ii)	Java Security Framework.	5
iii)	Indirect object reference in coding practice.	5
iv)	Attack vectors.	5
v)	Secure design steps.	5
vi)	Security architecture.	5

5-12-2023(M)


SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: <u>54</u>	Class: LY-Honours	Semester: VII (SVU 2020)
Programme: <u>Honours in Data Science and Analytics</u>		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What are the advantages of CNN (Convolutional Neural network) over ANN (Artificial neural network).	5
ii)	What is the benefit of using Momentum in Gradient Descent optimization?	5
iii)	Why Seq2Seq model is called as Conditional language model? Justify your answer.	5
iv)	Why Sigmoid or Tanh is not preferred to be used as the activation function in the hidden layer of the neural network?	5
v)	You are designing a deep learning system to detect driver fatigue in cars. It is crucial that your model detects fatigue, to prevent any accidents. Which of the following is the most appropriate evaluation metric: Accuracy, Precision, Recall, and Loss Value? Explain your choice.	5
vi)	Describe in short five phases of NLP	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Consider a neural network with three output neurons for a multiclass classification task. The network produces raw scores [2.0, 1.0, 0.5] as its output for a given input. Calculate the probabilities for each class using the softmax activation function. Explain how these probabilities can be interpreted in the context of classification.	5
ii)	Consider the Objective Function: $J(\theta) = (\theta - 4)^2$, $\theta_0 = 3$ (random initial value) $\alpha = 0.1$ (learning rate). Calculate values of θ_1 & θ_2 using gradient descent method.	5
OR		
Q2 A	Compare Deep Neural Networks (DNNs) with Shallow Neural Networks based on following parameters: i. Architecture Complexity ii. Feature Representation iii. Training challenges iv. Generalization Capability v. Computational efficiency	10

Q 2 B	Solve any One	10
i)	Name and explain any Four hyper-parameters used for training a deep neural network?	10
ii)	Explain k fold cross validation technique with example.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain forward and backward propagation in simple RNN with suitable mathematical equations.	10
ii)	Describe the role of the input gate, forget gate, and output gate in an LSTM network. How do these gates contribute to the network's ability to capture long-term dependencies?	10
iii)	Let us consider a Convolutional Neural Network having three different convolutional layers in its architecture as – Layer-1: Filter Size – 3 X 3, Number of Filters – 10, Stride – 1, Padding – 0 Layer-2: Filter Size – 5 X 5, Number of Filters – 20, Stride – 2, Padding – 0 Layer-3: Filter Size – 5 X 5, Number of Filters – 40, Stride – 2, Padding – 0 If we give the input image to the network of dimension 39 X 39 X 3, then determine the dimension of the vector after passing through a fully connected layer in the architecture.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Describe the architecture of a typical Seq2Seq model. Include the roles of the encoder and decoder, and explain how the model processes variable-length sequences.	10
ii)	Discuss one real-world application where Seq2Seq models are commonly employed. How does the architecture of Seq2Seq models make them suitable for this application?	10
iii)	Discuss the training process of the CBOW model in Word2Vec word embedding model. Include the roles of the input and output layers, the softmax function, and how word vectors are updated during training.	10

Que. No.	Question	Max. Marks
Q5	Write a short note on any Four	20
i)	Text pre-processing techniques (Any Two)	5
ii)	Adam Optimizer	5
iii)	Sentiment analysis	5
iv)	One hot encoding	5
v)	Cross-entropy loss	5
vi)	Different types of pooling in CNN.	5